AGRIVIVE / ERI-ROOT

TAMIL NADU / ERI-INSPIRED / SCREENED SUMP AND OUTLET

SERIES 23 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



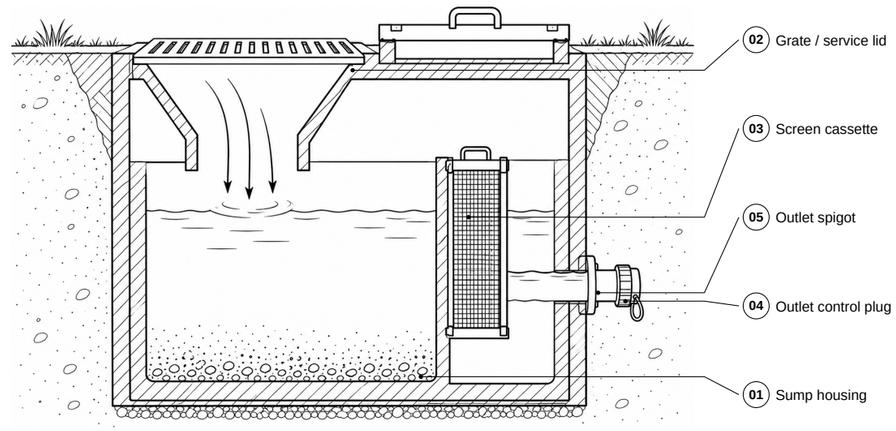
STATE CONTEXT / INDIGENOUS INSPIRATION

Tamil Nadu's eri tanks form longstanding irrigation storage and distribution systems. ERI-ROOT adapts interception and controlled outlet ideas into a small serviceable sump; it does not purify water.

Practice source: Government of India Kurukshetra: Eri systems | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 160 L x 100 W x 140 H

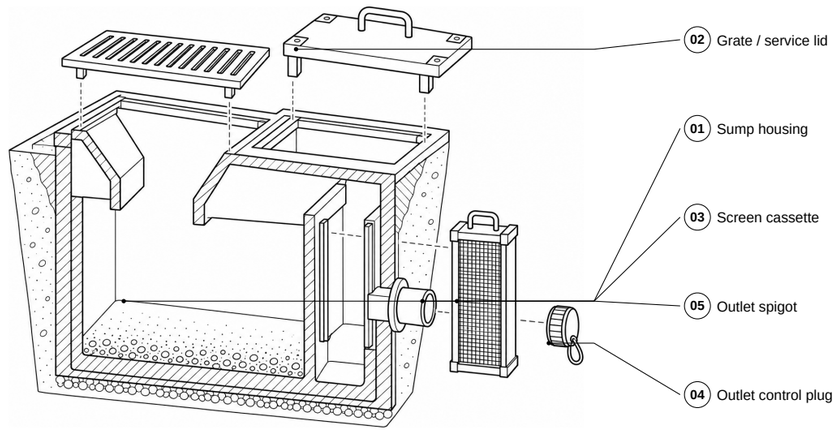


DESIGN / OPERATION NOTES

Set grate at runoff grade with accessible service clearance.
Fit screen and outlet-control plug in the gravity path.
Lift the cover to remove settled silt and clean the screen.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
160 L x 100 W x 140 H

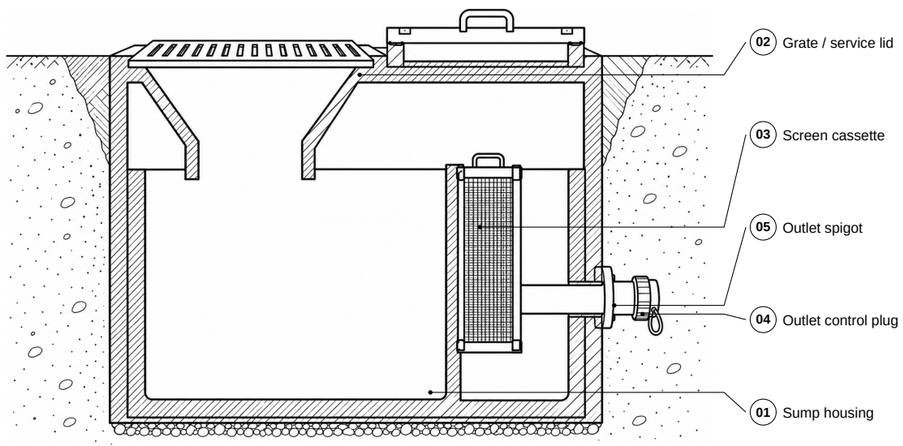


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

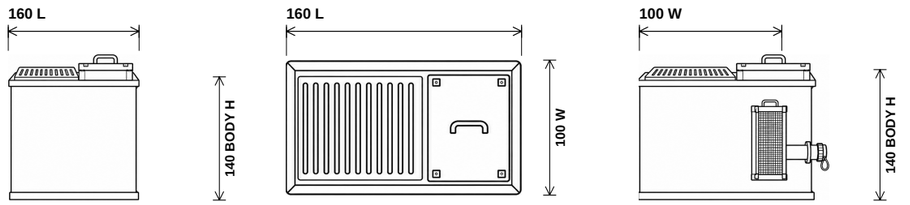
SECTION BASIS / mm
Sump 120 L x 80 W x 60 H; outlet25 ID



DESIGN / OPERATION NOTES

Fit screen and outlet-control plug in the gravity path.
Lift the cover to remove settled silt and clean the screen.

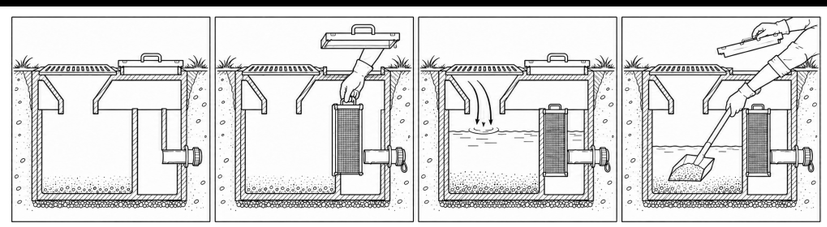
03 ORTHOGRAPHIC STUDIES / mm



FRONT / BODY **TOP / BODY** **SIDE SERVICE STUDY**

External 160 L x 100 W x 140 H. Gross sump 120 L x 80 W x 60 H; outlet DIA 25 ID.

05 INSTALLATION / OPERATING SEQUENCE



1 SEAT SUMP **2 FIT SCREEN** **3 CLOSE GRATE** **4 REMOVE SILT**

Prototype fit, access and operation require trials.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

External L x W x H	**160 x 100 x 140**
Sump L x W x H	**120 x 80 x 60**
Outlet ID	**25**

02 GEOMETRIC CHECK

GROSS SUMP ENVELOPE

576.0 mL

120 x 80 x 60 / 1000
Sediment reduces working volume; no purification claim

03 PARTS / QTY

01 Sump housing	**1**
02 Grate / service lid	**1 set**
03 Screen cassette	**1**
04 Outlet control plug	**1**
05 Outlet spigot	**Integral**

Body/grate/plug: candidate polymer. Screen and seal material TBD.

04 INTERFACES / CHECKS

Check plug retention and sediment bypass.

Test clogging, leakage and service access.

Measure retained volume with internals fitted.

05 DATUMS / DRAWING BASIS

A: grate plane. B: outlet invert. Preserve screen withdrawal clearance.
PRELIMINARY / VALIDATION PENDING

DESIGN DEVELOPMENT RECORD

01 / GEOMETRY
Nominal body sizes are proposed. Generated views are illustrative; small features and interfaces may vary.

02 / BEFORE FABRICATION
Develop one CAD model, assign wall thicknesses, tolerances and mating fits. Check print orientation and service access.

03 / TRIAL RECORD
Measure dimensions and retained volume or opening area. Record leakage, clogging and soil anchorage under observed conditions.

No discharge, treatment or durability performance is established.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-23 / P01	ERI-ROOT	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	23 / 28